MAST20030 Differential Equations
— Problem Sheet 5: Linear BVPs and PDEs —

1. Consider the following BVPs with "mixed" boundary conditions:

$$y''(x) + \lambda y(x) = 0, \quad y(0) = y'(L) = 0 \qquad L > 0.$$

   (a) Prove that the eigenvalues and eigenfunctions of the BVP are

$$\lambda_n = \frac{\pi^2(n-\frac{1}{2})^2}{L^2}, \qquad y_n(x) = \sin\frac{\pi(n-\frac{1}{2})x}{L}; \quad n = 1, 2, 3, \ldots$$

   (b) The eigenfunctions $y_n(x)$ are guaranteed to be orthogonal with respect to the inner product

$$\langle f, g\rangle = \alpha_L \int_0^L f(x)g(x)\,\mathrm{d}x,$$

   where α_L is an arbitrary positive constant. Determine α_L so that the eigenfunctions are orthonormal.

   (c) Suppose that a function f on $[0, L]$ may be written as a linear combination of eigenfunctions:

$$f(x) = \sum_{n=1}^{\infty} b_n y_n(x).$$

   Use orthonormality to derive a formula for the b_n.

   (d) Repeat the previous three parts for the "mixed" BVP

$$y''(x) + \lambda y(x) = 0, \quad y'(0) = y(L) = 0 \qquad L > 0.$$

2. State the order and the number of independent variables for the following PDE's. Decide whether the equation is linear or nonlinear, and homogeneous or inhomogeneous.

   (a) $\phi\dfrac{\partial^2\phi}{\partial x^2} + \dfrac{\partial\phi}{\partial y} = 1$

   (b) $\phi^2\dfrac{\partial^2\phi}{\partial x^2} + \dfrac{\partial^2\phi}{\partial y\partial z} = 1$

   (c) $\dfrac{\partial^4\phi}{\partial x^4} + \dfrac{\partial^2\phi}{\partial y^2} = 0$

   (d) $\dfrac{\partial^2\phi}{\partial x^2} + \dfrac{\partial^3\phi}{\partial y^2\partial z} = \phi^2$

3. Consider the heat equation

$$\frac{\partial\phi}{\partial t} = 4\frac{\partial^2\phi}{\partial x^2}$$

   for $\{(x, t) : 0 < x < 1, t > 0\}$, subject to the boundary and initial conditions

$$\begin{aligned} \phi(0, t) &= 0 \\ \phi(1, t) &= 0 \\ \phi(x, 0) &= \sin(3\pi x) \end{aligned}$$

(a) Using the method of separation of variables, show that the heat equation reduces to two ordinary differential equations (ODE's) of the form:

$$\begin{aligned} X''(x) - \lambda X(x) &= 0 \\ T'(t) - 4\lambda T(t) &= 0 \end{aligned}$$

where λ is the separation constant.

(b) By solving the ODE's in part (a) for the case $\lambda < 0$, determine the solution of the heat equation subject to the given initial and boundary conditions.

You may assume that solving the ODE's in part (a) for the cases $\lambda > 0$ and $\lambda = 0$ leads to trivial solutions. *You do not need to work through these two cases.*

4. (a) Determine the Fourier sine series for the function

$$f(x) = \begin{cases} 1 & 0 < x \leq \frac{L}{2} \\ 0 & \frac{L}{2} < x \leq L \end{cases}$$

(b) Using your answer to part (a), solve the diffusion equation

$$\frac{\partial \phi}{\partial t} = \frac{\partial^2 \phi}{\partial x^2}$$

for $\{(x,t) : 0 < x < L, \quad t > 0\}$ subject to the boundary conditions

$$\begin{aligned} \phi(0,t) &= 0 \\ \phi(L,t) &= 0 \\ \phi(x,0) &= f(x) \end{aligned}$$

5. (a) Determine the Fourier cosine series for the function

$$g(x) = x(L-x), \qquad 0 < x < L$$

(b) Using your answer to part (a), solve the diffusion equation

$$\frac{\partial \phi}{\partial t} = \frac{\partial^2 \phi}{\partial x^2}$$

for $\{(x,t) : 0 < x < L, \quad t > 0\}$ subject to the boundary conditions

$$\begin{aligned} \frac{\partial \phi}{\partial x}(0,t) &= 0 \\ \frac{\partial \phi}{\partial x}(L,t) &= 0 \\ \phi(x,0) &= g(x) \end{aligned}$$

6. Use separation of variables to find a solution the heat equation for $u(x,t)$, with $x \in [0,1]$ and $t > 0$, where the following boundary and initial conditions are imposed:

(a) $\partial_x u(0,t) = 0, \qquad \partial_x u(1,t) = 0, \qquad u(x,0) = 1 - x^2,$

(b) $u(0,t) = 0, \qquad u(1,t) = 0, \qquad u(x,0) = x\left(1 - \frac{1}{2}x\right).$

Can you show that your formal solutions are genuine?

7. Solve Laplace's equation

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0$$

for $\{(x, y) : 0 < x < L, 0 < y < L\}$, subject to the boundary conditions

$$\begin{aligned} \phi(x, 0) &= 0 \\ \frac{\partial\phi}{\partial x}(0, y) &= 0 \\ \frac{\partial\phi}{\partial x}(L, y) &= 0 \\ \phi(x, L) &= EL + \cos\left(\frac{\pi x}{L}\right) \end{aligned}$$

where E is a constant.

8. Solve Laplace's equation

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0$$

in the region

$$\{(x, y) : 0 < x < \pi,\ 0 < y < 1\}$$

subject to the boundary conditions

$$\phi(x, y) = \begin{cases} 0 & \text{at } x = 0 \\ 0 & \text{at } x = \pi \\ \sin x + \sin(2x) & \text{at } y = 0 \\ 2\sin x & \text{at } y = 1 \end{cases}$$

9. Solve Laplace's equation

$$\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}=0$$

in the region

$$\{(x,y):0<x<1,\,0<y<1\}$$

subject to the boundary conditions

$$\phi(x,y)=\begin{cases}0 & \text{at } x=0\\ 0 & \text{at } x=1\\ \sin(\pi x) & \text{at } y=0\\ \sin(2\pi x) & \text{at } y=1\end{cases}$$

10. Find solutions $u(x,y)$ to the (2D) Laplace equation, with $x,y\in[0,1]$, where the following boundary conditions are imposed:

(a) $u(0,y)=y(1-y)$, $u(1,y)=0$, $\partial_y u(x,0)=0$, $\partial_y u(x,1)=0$,

(b) $u(0,y)=0$, $u(1,y)=1-y^2$, $\partial_y u(x,0)=0$, $u(x,1)=x(1-x)$.

Are the formal solutions you obtained genuine solutions?

11. Find solutions $u(r,\theta)$ to the (2D) Laplace equation, *in polar coordinates*, on the given domain D with the following boundary conditions:

(a) $D=\left\{(r,\theta)\in[0,1]\times[0,\frac{\pi}{3}]\right\}$, $u(r,0)=0$, $u(r,\frac{\pi}{3})=0$, $u(1,\theta)=\theta(\frac{\pi}{3}-\theta)$,

(b) $D=\left\{(r,\theta)\in[0,1]\times[0,\frac{\pi}{2}]\right\}$, $\partial_\theta u(r,0)=0$, $\partial_\theta u(r,\frac{\pi}{2})=0$, $u(1,\theta)=\theta(1-\frac{3}{2\pi}\theta)$,

(c) $D=\left\{(r,\theta)\in[0,1]\times[0,\frac{\pi}{4}]\right\}$, $\partial_\theta u(r,0)=0$, $u(r,\frac{\pi}{4})=0$, $u(1,\theta)=1-\frac{9}{\pi^2}\theta$.

12. Consider the second order wave equation

$$\frac{\partial^2\phi}{\partial t^2}=36\frac{\partial^2\phi}{\partial x^2}$$

for $0<x<\pi$ and $t>0$, subject to the boundary and initial conditions

$$\begin{aligned}\phi(0,t) &= 0\\ \phi(\pi,t) &= 0\\ \phi(x,0) &= \sin(2x)\\ \frac{\partial\phi}{\partial t}(x,0) &= \sin x\end{aligned}$$

(a) Using the method of separation of variables, show that the wave equation reduces to two ordinary differential equations (ODE's) of the form:

$$\begin{aligned}X''(x)-\lambda X(x) &= 0\\ T''(t)-36\lambda T(t) &= 0\end{aligned}$$

where λ is the separation constant.

(b) By solving the ODE's in part (a) for the case $\lambda < 0$, determine the solution of the wave equation subject to the given initial and boundary conditions.

You may assume that solving the ODE's in part (a) for the cases $\lambda > 0$ and $\lambda = 0$ leads to trivial solutions. *You do not need to work through these two cases.*

13. Using separation of variables, solve the wave equation

$$\frac{\partial^2 \phi}{\partial x^2} - \frac{\partial^2 \phi}{\partial t^2} = 0$$

for $\{(x,t) : 0 < x < 1, t > 0\}$, subject to the boundary conditions

$$\begin{aligned} \phi(0,t) &= 0 \\ \phi(1,t) &= 0 \\ \frac{\partial \phi}{\partial t}(x,0) &= 0 \\ \phi(x,0) &= \sin(2\pi x) \end{aligned}$$

14. Find solutions $u(x,t)$ to the (1D) wave equation, with $x \in [0,1]$ and $t \geq 0$, where the following boundary and initial conditions are imposed:

$$\begin{array}{llll} u(0,t) = 0, & u(1,t) = 0, & u(x,0) = x(1-x), & \partial_t u(x,0) = x(1-x), \\ \partial_x u(0,t) = 0, & \partial_x u(1,t) = 0, & u(x,0) = 4x^3 - 6x^2 + 1, & \partial_t u(x,0) = 0, \end{array}$$

Repeat for the following variations on the wave equation:

- The *damped* wave equation: $\dfrac{\partial^2 u}{\partial t^2} = c^2 \dfrac{\partial^2 u}{\partial x^2} - \gamma \dfrac{\partial u}{\partial t}$, where $\gamma \geq 0$ is the *damping* coefficient.
- The *Klein-Gordon* equation of relativistic quantum mechanics: $\dfrac{\partial^2 u}{\partial t^2} = c^2 \dfrac{\partial^2 u}{\partial x^2} - \dfrac{m^2 c^4}{\hbar^2} u$, where m is the *mass*.
- The *telegraph* equation: $\dfrac{\partial^2 u}{\partial t^2} = c^2 \dfrac{\partial^2 u}{\partial x^2} - \gamma \dfrac{\partial u}{\partial t} - \zeta u$, where $\zeta \geq 0$ (see Wikipedia).

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— Solutions to Problem Sheet 5: Linear BVPs and PDEs —

1. (b) $\alpha_L = \dfrac{2}{L}$

   (c) $b_n = \dfrac{2}{L}\displaystyle\int_0^L f(x)\sin\frac{\pi\left(n-\frac{1}{2}\right)x}{L}\mathrm{d}x$

   (d) Same eigenvalues but eigenfunctions are $y_n(x) = \cos\dfrac{\pi\left(n-\frac{1}{2}\right)x}{L}$. The inner product that makes the eigenfunctions orthonormal is $\langle f, g\rangle = \dfrac{2}{L}\displaystyle\int_0^L f(x)g(x)\mathrm{d}x$. Any suitable function can be represented as $\displaystyle\sum_{n=1}^{\infty} a_n \cos\frac{\pi\left(n-\frac{1}{2}\right)x}{L}$ where $a_n = \dfrac{2}{L}\displaystyle\int_0^L f(x)\cos\frac{\pi\left(n-\frac{1}{2}\right)x}{L}\mathrm{d}x$.

2. (a) Second order, two independent variables, nonlinear.

   (b) Second order, three independent variables, nonlinear.

   (c) Fourth order, two independent variables, linear and homogeneous.

   (d) Third order, three independent variables, nonlinear.

3. (b) $\phi(x,t) = \mathrm{e}^{-36\pi^2 t}\sin 3\pi x$

4. (a) $s(x) = \dfrac{2}{\pi}\displaystyle\sum_{n=1}^{\infty}\frac{1}{n}\left(1-\cos\frac{n\pi}{2}\right)\sin\frac{n\pi x}{L}$

   (b) $\phi(x,t) = \dfrac{2}{\pi}\displaystyle\sum_{n=1}^{\infty}\frac{1}{n}\left(1-\cos\frac{n\pi}{2}\right)\mathrm{e}^{-\frac{n^2\pi^2}{L^2}t}\sin\frac{n\pi x}{L}$

5. (a) $s(x) = \dfrac{L^2}{6} - \dfrac{L^2}{\pi^2}\displaystyle\sum_{m=1}^{\infty}\frac{1}{m^2}\cos\frac{2m\pi x}{L}$

   (b) $\phi(x,t) = \dfrac{L^2}{6} - \dfrac{L^2}{\pi^2}\displaystyle\sum_{m=1}^{\infty}\frac{1}{m^2}\mathrm{e}^{-\frac{4m^2\pi^2}{L^2}t}\cos\frac{2m\pi x}{L}$

6. (a) $u(x,t) = \dfrac{2}{3} - \dfrac{4}{\pi^2}\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\mathrm{e}^{-\alpha n^2\pi^2 t}\cos n\pi x$

   (b) $u(x,t) = \displaystyle\sum_{n=1}^{\infty}\left(\frac{2\left(1-(-1)^n\right)}{n^3\pi^3} - \frac{(-1)^n}{n\pi}\right)\mathrm{e}^{-\alpha n^2\pi^2 t}\sin n\pi x$

   Both solutions are genuine solutions to the heat equation for $t > 0$ and $x \in [0,1]$.

7. $\phi(x,y) = Ey + \dfrac{\sinh\frac{\pi y}{L}}{\sinh\pi}\cos\dfrac{\pi x}{L}$

8. $\phi(x,y) = \left(\cosh y + \dfrac{2-\cosh 1}{\sinh 1}\sinh y\right)\sin x + \left(\cosh 2y - \dfrac{\cosh 2}{\sinh 2}\sinh 2y\right)\sin 2x$

9. $\phi(x,y) = \left(\cosh\pi y - \dfrac{\cosh\pi}{\sinh\pi}\sinh\pi y\right)\sin\pi x + \dfrac{\sinh 2\pi y}{\sinh 2\pi}\sin 2\pi x$

10. (a) $u(x,y) = \dfrac{1}{6}(1-x) - \dfrac{1}{\pi^2}\displaystyle\sum_{m=1}^{\infty}\frac{1}{m^2}\left(\cosh 2m\pi x - \frac{\cosh 2m\pi}{\sinh 2m\pi}\sinh 2m\pi x\right)\cos 2m\pi y$

(b)

$$u(x,y) = \frac{8}{\pi^3}\sum_{m=1}^{\infty}\frac{1}{(2m-1)^3}\frac{\cosh(2m-1)\pi y}{\cosh(2m-1)\pi}\sin(2m-1)\pi x - \frac{4}{\pi^3}\sum_{n=1}^{\infty}\frac{(-1)^n}{\left(n-\frac{1}{2}\right)^3}\frac{\sinh\left(n-\frac{1}{2}\right)\pi x}{\sinh\left(n-\frac{1}{2}\right)\pi}\cos\left(n-\frac{1}{2}\right)\pi y$$

11. (a) $u(r,\theta) = \dfrac{8}{9\pi}\displaystyle\sum_{m=1}^{\infty}\frac{1}{(2m-1)^3}r^{3(2m-1)}\sin 3(2m-1)\theta$

(b) $u(r,\theta) = \dfrac{\pi}{8} - \dfrac{1}{2\pi}\displaystyle\sum_{n=1}^{\infty}\frac{2-(-1)^n}{n^2}r^{2n}\cos 2n\theta$

(c) $u(r,\theta) = \displaystyle\sum_{n=1}^{\infty}\frac{18-(-1)^n(2n-1)\pi(4\pi-9)}{(2n-1)^2\pi^3}r^{4\left(n-\frac{1}{2}\right)}\cos 4\left(n-\frac{1}{2}\right)\theta$

12. (b) $u(x,t) = \dfrac{1}{6}\sin 6t\sin x + \cos 12t\sin 2x$

13. $u(x,t) = \cos 2\pi t\sin 2\pi x$

14. $u(x,t) = \dfrac{8}{\pi^3}\displaystyle\sum_{m=1}^{\infty}\frac{1}{(2m-1)^3}\left(\cos(2m-1)c\pi t + \frac{1}{(2m-1)\pi c}\sin(2m-1)c\pi t\right)\sin(2m-1)\pi x$

$u(x,t) = \dfrac{96}{\pi^4}\displaystyle\sum_{m=1}^{\infty}\frac{1}{(2m-1)^4}\cos(2m-1)c\pi t\cos(2m-1)\pi x$

For the three variations on the wave equation, solutions are similar but with the $T(t)$ functions respectively replaced with

$$T_n(t) = a_n e^{-\frac{1}{2}t\left(\gamma+\sqrt{\gamma^2-4c^2n^2\pi^2}\right)} + b_n e^{-\frac{1}{2}t\left(\gamma-\sqrt{\gamma^2-4c^2n^2\pi^2}\right)}$$

$$T_n(t) = a_n\cos\frac{c}{\hbar}t\sqrt{c^2m^2+\hbar^2n^2\pi^2} + b_n\sin\frac{c}{\hbar}t\sqrt{c^2m^2+\hbar^2n^2\pi^2}$$

$$T_n(t) = a_n e^{-\frac{1}{2}t\left(\gamma+\sqrt{\gamma^2-4c^2n^2\pi^2-4\zeta}\right)} + b_n e^{-\frac{1}{2}t\left(\gamma-\sqrt{\gamma^2-4c^2n^2\pi^2-4\zeta}\right)}$$